# Homework #9

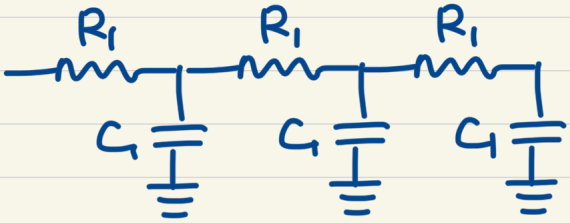
9-1 Consider a 5 mm-long, 4 λ-wide metal2 wire in a 0.6 μm process. The sheet resistance is 0.08 Ω/□ and the capacitance is 0.2 fF/μm. Construct 3-segment L-model and 3-segment π-model for the wire. (Please present models and calculate delay).

解. 0.6μm-CMOS, $\lambda = 0.3\mu m$

$L = 5mm$, $W = 4\lambda = 1.2\mu m$

$R_\square = 0.08\Omega/\square$, $C_l = 0.2fF/\mu m$

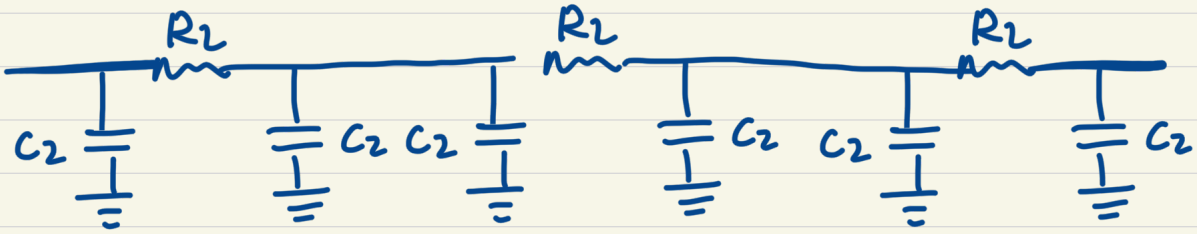
L-model




$R_1 = \frac{1}{3} R_\square \frac{L}{W} = 111.1\Omega$, $C_1 = \frac{1}{3} L C_l = 333.3fF$

$Delay = R_1 C_1 + 2R_1 C_1 + 3R_1 C_1 = 0.222ns$

π-model




$R_2 = \frac{1}{3} R_\square \frac{L}{W} = 111.1\Omega$, $C_2 = \frac{1}{6} L C_l = 166.7fF$

$Delay = 2R_2 C_2 + 4R_2 C_2 + 3R_2 C_2 = 0.167ns$

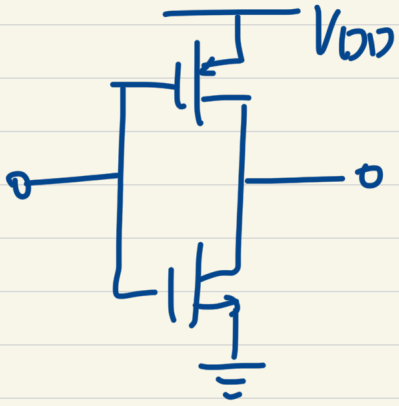
9-2 Assume an inverter is designed with a PMOS/NMOS ratio of 3.6 and the NMOS transistor is minimum size. (W = 0.375$\mu$m, L = 0.25$\mu$m, W/L =1.5). $VM$=1.25V, $VDD$=2.5V. Please describe the voltage transfer characteristic and noise margins of CMOS inverter. (use $V_{IL}$, $V_{IH}$, $NM_H$, $NM_L$ and g). The parameters shown in table 9.1.

Table9.1

| | $V_{T0}$ (V) | $\gamma$ ($V^{1/2}$) | $V_{DSAT}$ (V) | K ($\mu A/V^2$) | $\lambda$ ($V^{-1}$) |
|---|---|---|---|---|---|
| NMOS | 0.7 | 0.45 | 0.83 | 134 | 0.1 |
| PMOS | -0.8 | -0.4 | -1 | 50 | 0.2 |

$K = \frac{1}{2}\mu C_{OX}$

解.




Ratio=3.6 $(\frac{W}{L})_P = 3.4$

$W=0.375\mu m$, $L=0.25\mu m$, $\frac{W}{L}=1.5$

$V_{DD}=2.5V$, $V_M=1.25$

$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2(1+\lambda V_{out})$

$= k_n\frac{W}{L}(V_m-V_{T0})^2(1+\lambda_n V_m) = 68.40\mu A$

$g = \frac{1}{-I_D(V_m)} \times \frac{k_n(\frac{W}{L})_n V_{DSAT\,n} + k_p(\frac{W}{L})_p|V_{DSAT\,P}|}{\lambda_n+|\lambda_p|} = -21.29$

故 $V_{IH} = V_m - \frac{V_m}{g} = 1.31V$

$V_{IL} = V_m + \frac{V_{DD}-V_m}{g} = 1.19V$

$NM_H = V_{DD} - V_{IH} = 1.19V$

$NM_L = V_{IL} = 1.19V$

